MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6111WA

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Double triode, medium Mu, subminiature

Outline	---	3-1 (EIA)
Base	---	E8-10
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections

Pin No.	---	1	2	3	4	5	6	7	8
Element	---	2a	2g1	h	2k	1k	h	1g1	1a

ABSOLUTE-MAXIMUM RATINGS

Amplifier service:

Parameter:	Ef	Eb	eb	Ec1	Rk	Rg/g
Unit	V	Vdc	v	Vdc	Ohms	Meg
Maximum	6.6	250	360	---	---	1.1
Minimum	6.0	Note 8	---	-55	---	---
TEST CONDITIONS	6.3	100	---	0	220	---

Amplifier service:

Parameter:	Ehk	Ib/p	Ic/c	Pp/p	TE	Alt
Unit:	Vdc	mAdc	mAdc	W	°C	ft
Maximum:	+200	22	5.5	0.95	220	Note 14
Minimum:	-200	0.5	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	---

Pulse service:

Parameter	Ef	eb	Ec	egk	Rk/k	Ic/o
Unit:	V	v	Vdc	v	Ohms	mAdc
Maximum:	6.6	360	---	+50	---	5.5
Minimum:	6.0	---	-55	-100	---	---
TEST CONDITIONS:	6.3	150	---	+30 (see note 5)	1	---

Pulse service:

Parameter:	Pg/g	Pp/p	Du	tk/k	tp	ehk
Unit:	W	W	%	ms	uS	V
Maximum:	0.2	0.95	---	See figure 1	---	200
Minimum:	---	---	---	---	---	-200
TEST CONDITIONS:	---	---	1	---	10	---

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1301	Heater current	See note 1	0.4	II	If	265	---	315	mA
1236	Heater-cathode leakage	See note 1	1.0	II	Ihk	---	---	3.5	uAdc
1356	Total grid current	See notes 1 and 13	0.4	II	Ic1	0	---	-0.3	uAdc
1256	Electrode current (1) (anode)	See note 1	---	---	Ib	---	8.5	---	mAdc
			0.4	II	Ib	6.0	---	11.0	mAdc
1256	Electrode current (2) (anode)	Ec1 = -9.0 Vdc (see note 1)	0.4	II	Ib	---	---	100	uAdc
1306	Transconductance (1)	See note 1	---	---	Sm	---	5,000	---	umhos
			0.4	II	Sm	4,100	---	5,900	umhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---	
	Quality conformance inspection, part 2								
1211	Insulation of electrode	See note 1	2.5	I	R	250	---	---	Meg
1256	Electrode current (3) (anode)		2.5	I	Ib	---	---	1.5	mAdc
1306	Transconductance (2)	Ef = 5.7 V (see note 1)	2.5	I	ΔSm Ef	---	---	10	%
1256	Grid emission	Ef = 7.5 V; Eb = 250 Vdc; Ec1 = 20 Vdc; Rc/c = 1.0 Meg; Rk/k = 3,300 ohms (see notes 1 and 2)	2.5	I	Isc	0	---	-0.5	uAdc
1246	Audio frequency noise	Esig = 65 mVac; Rg = 0.1 Meg; Rk/k = 100 ohms; Ck = 100 uF; Rp = 0.01 Meg (see notes 3 and 4)	2.5	I	---	---	---	---	---
1316	Amplification factor	See note 1	6.5	S3	Mu	17	---	23	---
1256	Pulse cathode current (1)	Pulse service conditions (see notes 1, 5, and 6)	2.5	I	1k	310	---	---	ma
					1k	---	---	10	%
					Δtp				
1256	Pulse cathode current (2)	Pulse service conditions: Ef = 5.9 V (see notes 1, 5, and 6)	6.5	S3	1k	290	---	---	ma
1331	Direct-interelectrode capacitance	No shield (see note 1)	6.5	Code F	Cgp	1.2	---	1.8	pF
		No shield (see note 1)			Cin	1.4	---	2.4	pF
		No shield			Cout(1)	0.20	---	0.36	pF
		No shield			Cout(2)	0.22	---	0.42	pF
		No shield			Cgg	---	---	0.013	pF
		No shield			Cpp	---	---	0.50	pF
---	Tube operation time	See notes 1 and 7	4.0	S3	tt	---	---	20	sec
1031	High-frequency vibration	Rp = 10,000 ohms (see note 1)	2.5	I	Ep	---	---	20	mVac

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS BOGEY	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 - Continued								
1031	Sweep-frequency vibration	Rp = 10,000 ohms; 15 G; F = 70 to 2,000 Hz; t = 1 minute min, 3 minutes max; positions X and Y only (see notes 1 and 10)	6.5	Code G	ep	---	---	125	mVpp
1116	Lead fatigue		2.5	Code F	---	4.0	---	---	Arcs
1041	Shock	Ehk = +100 Vdc; 450 G (see note 11)	---	---	---	---	---	---	---
1031	Sweep-frequency vibration fatigue		6.5	Note 9	---	---	---	---	---
---	Post-shock and sweep-freq vibration fatigue test end points:								
1031	High-frequency vibration	Rc = 10,000 ohms	---	---	Ep	---	---	80	mVac
1338	Heater cathode leakage	See note 1	---	---	Ihk	---	---	7	µAdc
1308	Transconductance(1) change of individual tubes		---	---	ΔSm Ef	---	---	15	%
1244	Total grid current	See notes 1 and 13	---	---	Icl	0	---	-1.0	µAdc
2126	Envelope strain		2.5	I	---	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---	---

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: 1ST SAMPLE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC: COMBINED SAMPLES	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3									
1506	Heater-cycling life	Ef = 7.0 V; Eb = Ecl = 0; Ehk = 140 Vac; 1 min on, 4 min off	---	---	---	---	---	---	---	---
1516	Stability life	Ehk = +200 Vdc; Rg/g = 1.0 Meg; TA = room	1.0	I	---	---	---	---	---	---
---	Stability life-test end points:									
1308	Transconductance (1) change of individual tubes		---	---	---	---	ΔSm t	---	10	%
1501	Intermittent life (1)	Eb = 250 Vdc; Ecl = 0; Rk/k = 3,300 ohms; Ehk = +200 Vdc; Rg/g = 1.0 Meg; TE = 130°C (see note 12)	---	---	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMITS		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 3 - Continued									
---	Intermittent life - test (1) end points: (500 hours)		---	---	1	3	---	---	---	---
---	Inoperatives		---	---	1	3	Ic1	0	-0.9	µAdc
1268	Total grid current	See notes 1 and 13	---	---	1	3	If	276	324	mA
1301	Heater current		---	---	1	3	ΔSm t	---	20	%
1306	Transconductance (1) change of individual tubes		---	---	1	3	ΔSm Ef	---	20	%
1306	Transconductance (2)		---	---	1	3	Ihk	---	10	µAdc
1338	Heater-cathode leakage	See note 1	---	---	1	3	R	50	---	Meg
1211	Insulation of electrodes		---	---	3	6	---	---	---	---
---	Total defectives		---	---	---	---	---	---	---	---
1501	Intermittent pulse life	Et = 250 Vdc; Ec1 = 25 Vdc; Rk/k = 0; Ri/p = 330 ohms; eg1 = +30±1 v. TA = room (see notes 4 and 5)	---	---	---	---	---	---	---	---
---	Intermittent pulse life-test end points (500 hours									
---	Inoperatives		---	---	1	3	---	---	---	---
---	Pulse emission (1)		---	---	1	3	Ik	200	---	ma
---	Pulse emission (1), change of individual tubes		---	---	1	3	Δik	---	-35	
---	Total defectives		---	---	2	6	Δik	---	+50	
1501	Intermittent life (2)	Stability life test, TE = +220°C (min) (see note 12)	---	---	---	---	---	---	---	---
---	Intermittent life-test (2) end points: (500 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1268	Total grid current	See notes 1 and 13	---	---	1	3	Ic1	0	-0.7	µAdc
1301	Heater current		---	---	1	3	If	276	324	mA
1306	Transconductance (1) change of individual tubes		---	---	1	3	ΔSm t	---	20	%
1306	Transconductance (2)		---	---	1	3	ΔSm Ef	---	15	%
1338	Heater-cathode leakage	See note 1	---	---	1	3	Ihk	---	10	µAdc
1211	Insulation of electrodes		---	---	1	3	R	50	---	Meg
1306	Transconductance (1) average change		---	---	---	---	Avg ΔSm	---	15	%
---	Total defectives		---	---	3	6	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC		SYMBOL	LIMIT		UNIT
					1ST SAMPLE	COMBINED SAMPLES		MIN	MAX	
	Quality conformance inspection, part 2 -Continued									
---	Intermittent life-test (2) end points: (1,000 hours)									
---	Inoperatives		---	---	1	3	---	---	---	---
1266	Total grid current	See notes 1 and 13	---	---	1	3	Ic1	0	-0.9	uAdc
1301	Heater current		---	---	1	3	If	275	328	ma
1306	Transconductance (1), change of individual tubes		---	---	1	3	ΔSm 1	---	25	%
1306	Transconductance (2)		---	---	1	3	ΔSm Ef	---	20	%
1336	Heater-cathode leakage	See note 1	---	---	1	3	Ihk	---	10	uAdc
1211	Insulation of electrodes		---	---	1	3	R	25	---	Meg
---	Total defectives		---	---	4	8	---	---	---	---

NOTES:

1. Test each unit separately.

2. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions specified below. A 3-minute test shall not be permitted. Test at specified conditions immediately after preheating without removing the tube from the socket. Transferring the tube under test from a preheat socket to a test socket or removal of applied voltages (except to change bias) between preheat and test is not permitted. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef	Eb	Ec1	Rg	Rk/k
V	Vdc	Vdc	Meg	Ohms
7.5	250	0	1.0	3,300

3. Tie 1k to 2k; 1g to 2g; and 1a to 2a.

4. The rejection level shall be set at the VU meter reading obtained during calibration.

5. The grid pulse shall be a square wave meeting the pulse shape requirements of method 1296 and in addition the maximum amplitude shall occur within the first 20 percent of tp; tp = 10 µs; duty cycle 1.0 percent; egk = +30 v; the pulse shall be applied to the grid by means of a driving circuit which produces the specified peak pulse voltage directly at the grid terminals with respect to the cathode. Grid resistance, not exceeding 50 ohms may be inserted to prevent oscillation providing readjustment of grid drive is made to maintain the specified pulse amplitude directly at the grid terminal. The dc bias shall be equal to or greater than -25 volts.

6. Pulse cathode current shall be measured by means of a high-impedance oscilloscope, or equivalent device, connected across a 1.0±0.01 ohm cathode resistor. The specified limit refers to the maximum of the pulse amplitude. The variation of the output pulse amplitude between 20 and 80 percent tp shall not exceed the specified limits where tp is defined as in method 1296. Peak cathode current shall be read after 10 seconds, or in the case of slumping peak cathode current, when a stable reading is obtained.

7. Tube operation time is the time in seconds required for the anode current to attain 90 percent of the 2-minute anode current value measured at test conditions specified below. The tube filament shall be energized instantaneously and the tube shall have had no potentials applied for a period of at least 1 hour prior to the performance of this test. The total filament supply impedance of the transformer and series external resistance shall be 2.6 ohms ±20 percent. The applied filament voltage shall be adjusted so that the heater voltage listed under test conditions is the actual voltage applied to the tube under test terminals for a bogey tube, or equivalent.

Ef	Eb	Ec1	Rk/k
V	Vdc	Vdc	Ohms
6.3	100	0	220

8. The simultaneous use of absolute maximum dissipation and envelope temperature requires anode voltage derating below the absolute maximum rating. The extent of control under simultaneous absolute maximum element voltage and wattage is exemplified by intermittent life test (1) conditions.

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9. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. When one lot has passed, the 6-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter F, shall apply.

10. The specified voltages shall be applied to the tube during vibration. The value of Eb under test conditions shall be regarded as Ebb, and shall be applied to the tube through the specified resistor, (Rp). The voltage (ep), produced across the resistor, (Rp), as a result of vibration, shall be measured with a suitable recording device. This device shall have an appropriate voltage range and shall have the ability to indicate with an error of less than 10 percent, the peak to peak value of a sine wave at all frequencies from 50 to 20,000 Hz. A Bruel and Kjaer recorder type 2304 with potentiometer type 2347 suitably calibrated, or an equivalent device, may be used.

    The voltage (ep) produced across the resistor (Rp) as a result of vibration shall be coupled to a compensating amplifier and low-pass filter. The coupling capacitor shall have a minimum value of 0.25 µF and the input to the amplifier shall have a high impedance (1 megohm or more) and the amplifier shall be adjusted to compensate for any insertion losses in the passband of the filter. The shunting capacitance of the cabling and amplifier input shall be not greater than 300 pF. The combined frequency response of the amplifier and filter shall be flat within ±1 dB from 20 to 10,000 Hz as referenced at 1,000 Hz shall be down 3 ±1 dB at 12,000 Hz, down to ±1 dB at 15,000 Hz, and down 20 ±1 dB at 20,000 Hz. The impedance of the anode and screen voltage supplies shall not exceed that of a 40 µF capacitor at 10 Hz.

11. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.

12. Envelope temperature (TE) requirements, when measured in accordance with the temperature by loop-thermocouple measurement (method 1226) will be satisfied if a tube having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any positions on the life-test rack.

13. This test shall be performed at the conclusion of the holding period.

14. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

15. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

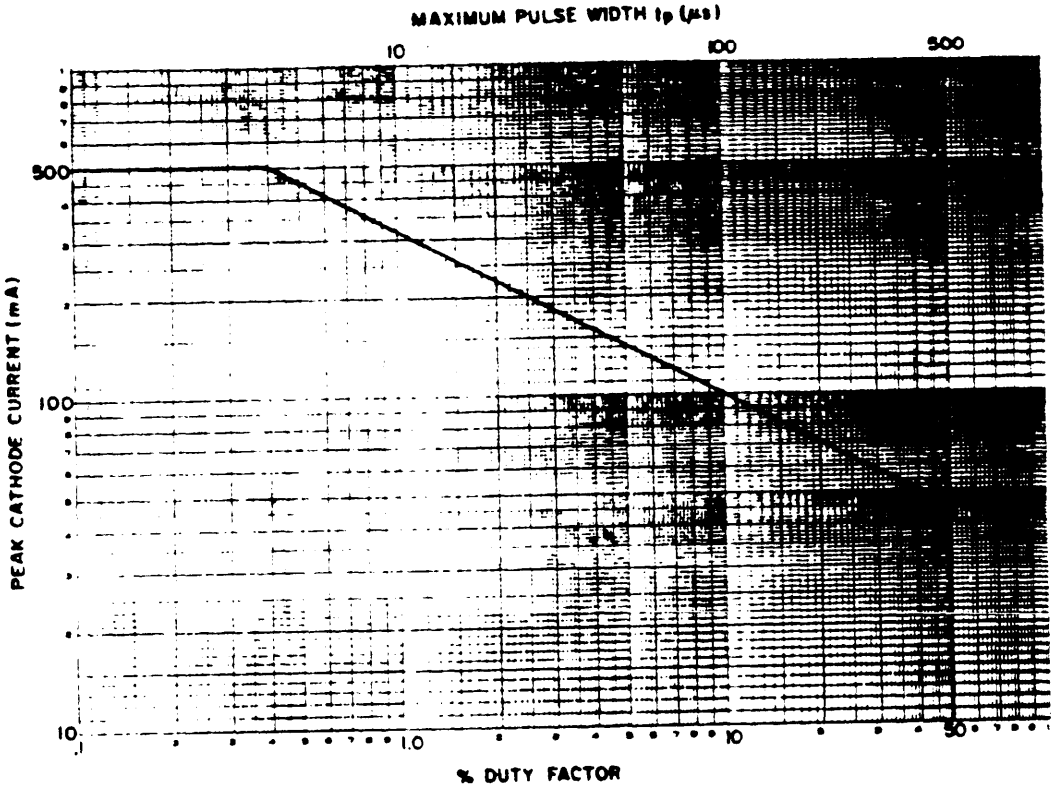
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Area of permissible operation/section

NOTES:
1. The area below the line establishes permissible peak cathode current per section for duty factors up to 50%. The rating chart shall not be employed to estimate pulse capabilities at duty factors above this value.
2. Duty factor is defined as the ratio of the average current to the maximum peak current occurring in any 1000 us period.
3. 500 milliamperes is the absolute maximum current at any duty factor less than 0.1%.

FIGURE 1. Pulse rating chart for tube type 6111WA.

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